

A flexible F2 topside inverse model

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Summary

This prototype fits **foF2**, **hmF2**, and **topside shape independently** instead of restricting candidates to a shifted, stretched prior. One branch uses four shape components learned from 100 global IRI-2020 profiles; a second uses a generalized Chapman F2 layer. The code tests both branches with O-mode and X-mode return ranges and selects 3-D candidates using all accepted O/X returns. The 1 km homing gate and 2–10 MHz sweep at 0.1 MHz spacing are unchanged.

The shape test on eight separate IRI-2016 profiles is strong: the IRI basis has **4.8 km mean absolute hmF2 error** and **0.11% peak-normalized topside RMS** when both truth and retrieval use a scalar vertical ray operator. A single Chapman layer gives 18.1 km and 11.16% on these profiles. The comparison shows that one width parameter was too restrictive. The full 3-D O/X check is reported separately below; the scalar numbers do not measure magnetoionic ray-tracing accuracy.

Data and independence

The basis was trained on IRI-2020 density sampled at 5 latitudes, 5 longitudes, and 4 seasons/local times in 2011. F10.7 was fixed at 120. Each profile was aligned at its F2 maximum, divided by its peak density, and sampled every 5 km from the peak to 600 km above it. The validation ionograms and density grid come from a separately generated IRI-2016 case at 2010-01-01 12:00 UTC. The fitting stage reads only its accepted-return file and an independent PyIRI background for horizontal variation. It does not read the IRI-2016 density grid; density is opened during evaluation.

IRI-2016 and IRI-2020 are related versions of one empirical model, so this is a held-out code-version and location test, **not an independent model-family test**. The 3-D truth and candidate ionograms share the ray tracer, fan, and homing code. The scalar tests use a separate tabulated-profile integrator but the same vertical cold-plasma equation. No measured ionogram is used.

Retrieval method

For the IRI branch, the log density above the peak is

$$\log[N(h_m + s)/N_m] = \mu(s) + \sum_{k=1}^4 a_k e_k(s), \quad N_m = (f_o/0.00898)^2 \text{ cm}^{-3}.$$

Here μ is the mean normalized IRI profile and e_k are its first four principal components, scaled so $a_k = 1$ is a one-standard-deviation change. The independent fitted parameters are f_o , h_m , and the four a_k . A 55 km Chapman bottomside supplies density below the peak; that region is weakly constrained by topside reflections. The competing Chapman branch fits its own topside scale height and curvature.

Search and forward checks

The fast proposal stage computes vertical two-way group range by quadrature through each candidate density. It fits per-frequency median O/X ranges with a robust loss, a soft 100 kHz nose constraint, and weak shape priors. The scalar X operator has nuisance frequency and range offsets because it omits magnetoionic propagation; those offsets are not interpreted as plasma parameters. The highest return bin is excluded from the fast range residual because group delay is singular at the limiting frequency. For full-ray checks, each column receives a tunable fraction g of PyIRI’s smooth horizontal peak-frequency and peak-height variation. We estimate PyIRI’s local peak with a three-point quadratic and set $f_o(x) = f_o(x_s)[f_{o,\text{prior}}(x)/f_{o,\text{prior}}(x_s)]^g$ and $h_m(x) = h_m(x_s) + g[h_{m,\text{prior}}(x) - h_{m,\text{prior}}(x_s)]$. The tested gains are 0, 0.15, 0.30, 0.5, and 1. Candidate density is then traced by the option-D 3-D O/X fan. One bounded height refinement converts the median shared-frequency O/X group-range residual to an approximate reflector-height step using $\partial R/\partial h_m \simeq -2$; both 75% and 100% steps receive full-ray checks. Final selection uses the existing score: 55% symmetric accepted-return distance, 25% last-return-frequency error, and 20% common-frequency median-range error. **Every accepted return** enters the full-ray distance score.

Shape and family validation

Truth family / scalar test	Selected branch	hmF2 MAE	foF2 MAE	Mean topside RMS
Eight IRI-2016 profiles	IRI basis, 8/8	4.80 km	0.0069 MHz	0.11%
Same eight, forced single Chapman	Chapman	18.11 km	0.0400 MHz	11.16%
Four analytic Chapman profiles	Chapman, 4/4	3.15 km	0.0203 MHz	0.23%
Same four, forced IRI basis	IRI basis	31.04 km	0.0663 MHz	8.45%

The analytic Chapman cases span 45–95 km topside scales and one curved tail. The family choice above is made from range residuals, not from truth density. These are idealized scalar ionograms; they establish basis flexibility and family selection, then the full 3-D case tests transfer to the actual O/X generator.

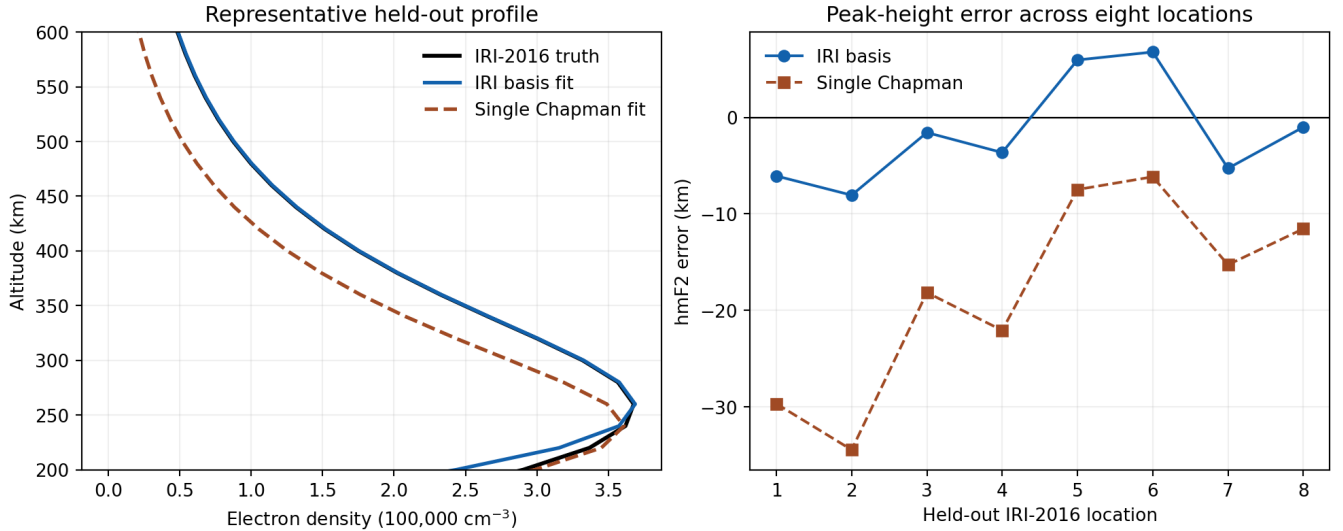


Figure 1: A held-out IRI-2016 density cut (left) and the hmF2 errors at all eight validation locations (right). The IRI-derived basis follows the topside much more closely than the single Chapman branch. The divergence below the peak is outside the well-observed topside region.

Full 3-D O/X check

The held-out IRI-2016 ionogram contains 56 O and 60 X accepted returns. Candidates were traced through complete 3-D density grids and ranked by the O/X ionogram score, before opening the truth density. The score combines symmetric distance between all accepted returns (55%), last-return frequency (25%), and median range at common frequencies (20%). Smaller is better; it is a dimensionless selection score, not a density-error estimate.

Candidate	Full-ray score	foF2 error	hmF2 error	Topside RMS
Selected: IRI basis, zero gradient, full height step	0.0881	−0.0065 MHz	+4.26 km	0.24%
IRI basis, 0.15 gradient, full height step	0.0887	−0.0062 MHz	+4.95 km	0.11%
IRI basis, zero gradient, 75% height step	0.1183	−0.0085 MHz	+2.19 km	0.81%
Previous shifted-PyIRI retrieval	0.1498	—	—	—
IRI basis, 0.15 gradient, before height step	0.1865	−0.0120 MHz	−4.85 km	2.57%
Single Chapman candidate	0.2119	−0.0596 MHz	−37.11 km	18.73%

Density errors in this table are **diagnostics after selection**. The selected profile has foF2 5.4015 MHz versus 5.4080 MHz in truth, and hmF2 261.55 km versus 257.29 km. The peak-density error is approximately −0.24%, from the squared foF2 ratio. Topside RMS is the density RMSE from truth hmF2 to 600 km, divided by truth peak density. The selected profile’s broader 150–600 km RMS is 10.35%, exposing the assumed 55 km Chapman bottomside; the vertical topside ionogram does not constrain that region well. The 0.15-gradient fit has a marginally higher ionogram score yet a slightly smaller topside RMS, so these two candidates are not meaningfully separated by this one case.

The selected O-mode nose is 5.4 MHz, matching truth. Its X-mode last return is 5.7 MHz versus 5.4 MHz in truth. The remaining nose and high-range differences in the figure are real residuals, even though the overall score improved by 41% against the previous retrieval. The Chapman candidate’s plausible 0.212 ionogram score but 37 km peak-height error illustrates the shape/height ambiguity that motivated the IRI-derived basis.

The figure places every accepted O and X return at its exact 100 kHz frequency and rounded 1 km group-range bin. Both modes are overlaid in each ionogram; the density cut is at the sounder position. Peak frequencies and heights use a three-point quadratic around the maximum of the 20 km forward grids.

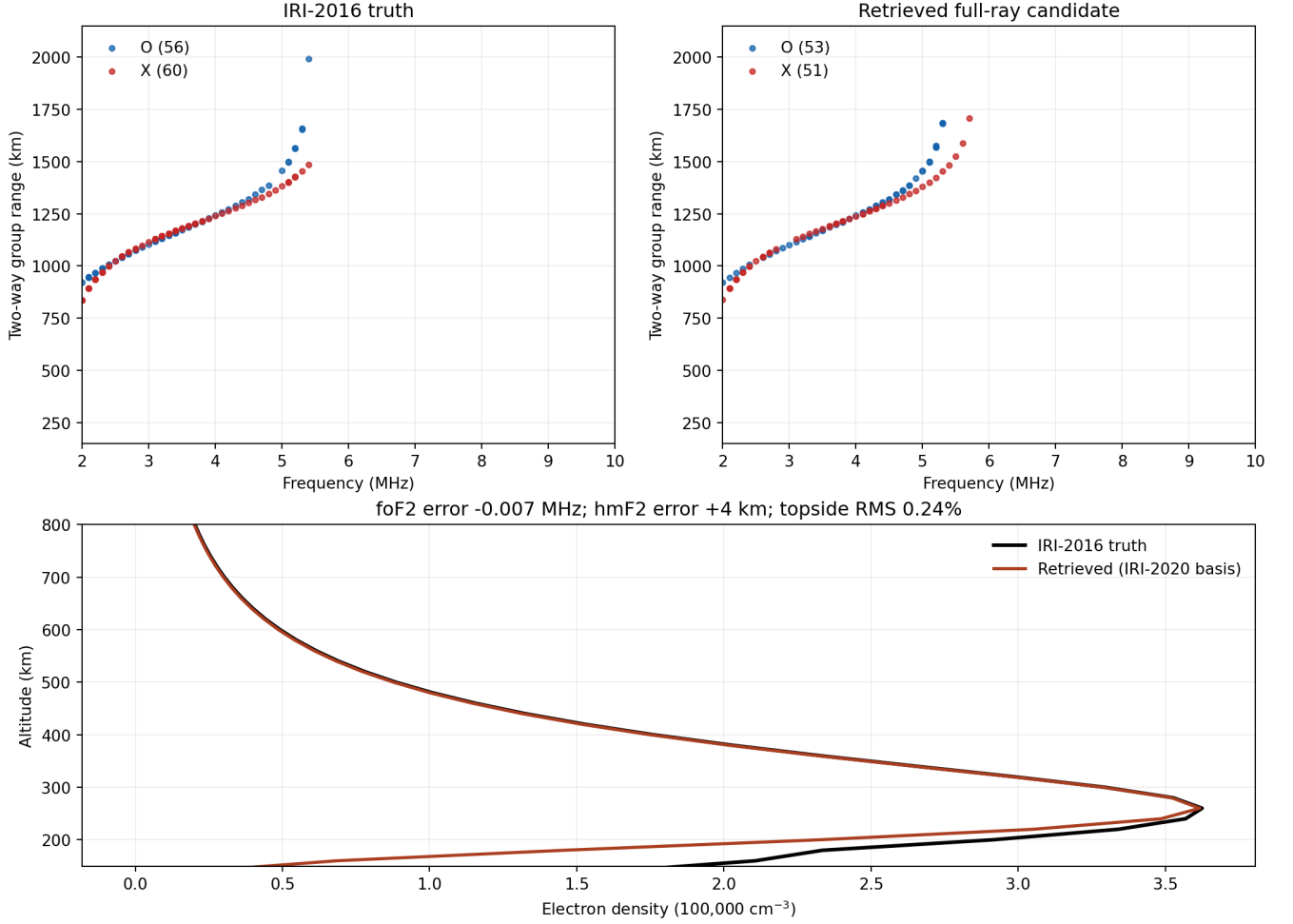


Figure 2: IRI-2016 truth and the minimum-score full-ray candidate. Blue is O mode and red is X mode; all accepted returns are plotted. The lower panel compares local density profiles.

Limits and next step

The inverse still has limited bottomside information and can trade local shape against 3-D refraction. The scalar X offsets are proposal-stage corrections, not a substitute for the full magnetoionic solver. The current comparison is exploratory because the new model family was developed after inspecting an earlier case at this location. A fresh location and epoch, held out before method tuning, are needed for a prospective accuracy claim. Joint vertical and oblique fitting across multiple sounder positions should help separate peak height from horizontal gradient and topside curvature.